Suyash Chaudhary

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.48
Chennai Public School, Thirumazhisai	Class XII (CBSE)	2018	87.60%
Chennai Public School, Thirumazhisai	Class X (CBSE)	2016	9.30%

EXPERIENCE

Teaching Assistant · IIT Hyderabad | Discrete Mathematics · Compilers I · GPU Programming

- Conducted introductory and practice sessions, graded assignments and quizzes, and **mentored course projects**.
- Built a **LeetCode-style automated coding-assessment** environment on the lab's servers to run **secure in-lab exams**.

Founding Engineer · Aware Precision Medicine Pvt. Ltd. | Flutter · Rails · AWS · Node

- Led a **lean engineering team** end-to-end -- sprint planning, task delegation, and progress tracking through to release.
- Built a **multi-vendor lab-testing marketplace** linking pan-India labs to users by serviceable pincode, with in-app booking, a **recommendation engine** suggesting tests from tracked vitals, and webhook-driven order and lab-status updates.
- Set up an **OCR document-processing pipeline** automating lab onboarding by digitizing test catalogs and price lists, and extracting structured results from scanned patient reports for downstream vitals tracking and recommendations.
- Ingested wearable biometrics through a **stream-processing pipeline** (Redis queues, Sidekiq workers) into a **heart-score engine** fusing lab vitals with sleep/activity/heart-rate signals via time-series moving averages on scheduled jobs.
- Shipped a **daily-task tracker** with medication reminders and push notifications, and a **gamified rewards engine** with global/group leaderboards; instrumented app-wide event tracking via **Google Analytics and Mixpanel**.

SKILLS

Languages: C++ · Python · JavaScript · TypeScript · CUDA · Dart · Ruby · C · SQL · Bash

Libraries: React · Node.js · Express · Flutter · Ruby on Rails · PyTorch · Numpy · Pandas

Tools: AWS · Docker · Git · Linux · LLVM · CMake · Sidekiq · Redis · Nsight Compute · PostgreSQL · Nginx

Domains: Full-Stack Web · Cross-Platform Mobile · ML Compilers · Parallel & GPU Programming · Static Program Analysis · Compiler Design · REST APIs · Networking · Security · Machine Learning · Systems Programming

Fundamentals: Data Structures & Algorithms · Operating Systems · Computer Networks · DBMS · Computer Architecture

RELEVANT COURSES

Inference, Program Analysis & Compiler Optimizations · GPU Programming · Foundations of Machine Learning · Fraud Analytics · Visual Computing · Wireless Networks & Security · Deep Learning Specialization - DeepLearning.AI

PROJECTS

CUDA Kernel Optimization Using Warp Specialization | C++, CUDA, GPU Programming, Optimization · Github

- Implemented **warp specialization** in GPU kernels to overlap memory transfers with computation and **raise throughput**.
- Applied **producer/consumer scheduling** to GEMM (FP32, FP64 & Tensor), Stencil, 3D Convolution, and 2D Jacobi kernels; profiled with **NVIDIA Nsight Compute** for memory, cache, and hardware-utilization characteristics.
- **Benchmarked on PolyBench** against a suite of optimizations -- bank-conflict-free shared-memory tiling, double

buffering, CUDA/Tensor-Core overlap, async memory operations, and named-barrier scheduling.
- **Cut kernel runtime** by up to **~60% (2.5× speedup)** over naive baselines on 4096×4096 workloads.

Tensor Dimension Equality Analysis | Python, Static Analysis, SMT, ML Compilers · Github

- Built a **flow-sensitive, path-sensitive** static analysis pass that proves whether two tensor dimensions are equal.
- Modeled **PyTorch** tensor shapes across broadcasting and shape-preserving operations as **symbolic dimension constraints**.
- Translated the Python AST into a control-flow graph, applied SSA renaming, and ran a **worklist algorithm** over abstract states; resolved equality queries via the **Z3 SMT solver** under per-path constraints.

Alias Analysis for LLVM IR | C++, LLVM, Data-flow Analysis, Compilers · Github

- Built a compiler analysis pass that conservatively determines whether two pointers in a program **may alias**.
- Implemented a **flow-sensitive, intra-procedural** dataflow analysis over **LLVM IR** to model pointers' memory locations.
- Used a **worklist algorithm** with **Sparse Bit Vectors** to represent points-to sets, with strong/weak update support.

Automated WPA3 Downgrade Attack (Dragonblood) | Python, Network Security, 802.11 · Github

- Built an automated tool to exploit the **WPA3 transition-mode downgrade** vulnerability and capture the WPA2 handshake.
- Parsed 802.11 RSN elements to find **vulnerable transition-mode APs**; deployed a **rogue AP to downgrade clients** to WPA2.
- Hand-parsed 802.11 RSN/AKM fields with **Scapy**; discovered APs via **airodump-ng**; ran a rogue AP with **hostapd-mana** and deauthentication to capture the WPA2 4-way handshake for offline analysis.
- Demonstrated end-to-end as an **authorized proof-of-concept** on the campus network.

Graph Compression using Neighborhood Similarity | C++, Graph Algorithms, Compression · Github

- Built novel compression algorithms to reduce the **memory footprint** of large real-world graphs.
- Designed and implemented **lossless and lossy compression** algorithms; Extended common graph queries to **run directly on the compressed graph**; Benchmarked on large real-world social, road, and collaboration networks.
- Utilized **neighbour-similarity guided node fusion** with query-driven selective un-fusion to keep query results exact.
- Compressed real-world SNAP graphs by **up to 65%** (~37% average), beating the **GraphZip baseline by 10%+**.

HOBBIES

Gym · Travel · Books · Cycling · Playing Badminton · Watching Cricket · Video Games · Anime · Driving